Foundations of Bayesian Statistics, and Inference

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Mathematical Methods for Physicists II

Coursework Notes and Computational Guides

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This guide builds on probability and likelihood from Sections 1–4 and explains the methods shared by Notebooks 5A and 5B.

1 From likelihood to posterior

Section 4B used a likelihood to compare candidate decay constants. Maximum likelihood selects its peak. Bayesian inference adds a prior and produces a distribution over parameter values.

$$p(\theta \mid D, H) = \frac{p(D \mid \theta, H)p(\theta \mid H)}{p(D \mid H)}, \qquad p(D \mid H) = \int p(D \mid \theta, H)p(\theta \mid H)\, d\theta.$$

Read the bar $\mid$ as "given." Here H is the model and its assumptions, D is the data we observed, and θ is what we want to learn.

1.1 A one-flip example

Suppose there are two coins: a fair coin with heads probability $\theta = 0.50$ and a biased coin with $\theta = 0.75$. We pick one at random, flip it once, and get heads.

- **Prior,** $p(\theta \mid H)$**:** Before the flip, each coin has a probability of 0.50 of being picked.
- **Likelihood,** $p(D \mid \theta, H)$**:** Given the coin, how likely was the observed head? The answer is 0.50 for the fair coin and 0.75 for the biased coin. The data stay fixed while we change which coin we consider.
- **Evidence,** $p(D \mid H)$**:** What was the overall chance of heads before we knew which coin we had? Average the two likelihoods using their prior weights:

$$p(D \mid H) = 0.50(0.50) + 0.50(0.75) = 0.625.$$

- **Posterior,** $p(\theta \mid D, H)$**:** Now that we saw heads, how likely is each coin? For the biased coin,

$$p(\theta = 0.75 \mid D, H) = \frac{0.75 \times 0.50}{0.625} = 0.60.$$

  The fair coin has posterior probability 0.40. One head makes the biased coin more plausible, but either coin could have produced it.

The evidence normalizes the prior times likelihood weights so they add to one. When θ can vary continuously, the sum becomes the integral above, and probabilities are areas under the density curve. The next example calls the unknown heads probability p and uses a Beta distribution to describe which values of p are plausible. Its two parameters, a and b, control the shape of that curve.

A uniform prior depends on the variable chosen: a prior uniform in temperature is not uniform in inverse temperature.

2 A worked example: update a coin model

A Beta(a, b) prior is proportional to $p^{a-1}(1-p)^{b-1}$. With h heads in n independent flips at the same p, the likelihood is proportional to $p^h(1-p)^{n-h}$. The binomial coefficient is fixed by the observed counts and cancels when normalizing the posterior.

$$p \mid D \sim \text{Beta}(a + h, b + n - h) = \text{Beta}(10, 6) \quad (a = b = 2,\ h = 8,\ n = 12).$$

This is a conjugate update because the posterior stays in the Beta family. The posterior mean, 0.625, lies between the prior mean 0.5 and the observed fraction 8/12. The central 95% credible interval is about [0.384, 0.837], leaving 2.5% posterior probability in each tail. A frequentist confidence interval instead describes the coverage of a procedure across repeated samples.

Notebook 5A repeats the update on a grid and overlays the analytic density. It then predicts the next flip by averaging its heads probability over the posterior, giving 0.625. In general prediction averages over parameter uncertainty and includes the randomness of new observations.

3 Build and read a grid posterior

1. Split the prior ranges into cells. Evaluate prior density and likelihood at cell midpoints; multiply by the cell width or area to approximate probability mass.
2. Add log prior and log likelihood. Subtract the largest log weight before exponentiating, then divide by the sum of weights. Equal cell areas cancel; unequal cells require their own areas.
3. To average out an unknown parameter, sum along its axis. This is marginalization. A slice at its best value is a different calculation.
4. Use a mass-weighted average for the mean. Read equal-tail intervals from cumulative mass at cell edges, interpolating within cells. Divide by cell width when plotting a one-dimensional density.

The MAP estimate maximizes posterior density and can differ from the mean or the maximum-likelihood estimate. Combining two marginal means yields one representative pair, but the prediction at that pair need not equal the posterior average prediction. Check the prior boundaries and refine the grid if the result changes with resolution.

4 Gaussian noise and model checks

For independent measurements y_i with known noise standard deviation σ and model mean $m_i(\theta)$, the log likelihood is $-\frac{1}{2}\sum_i (y_i - m_i)^2/\sigma^2 + C$. The constant is common to candidates here. If the noise scale is unknown or differs across models, keep any normalization term that changes. A residual plot checks for systematic mismatch. A posterior predictive check would draw parameters, simulate data including noise, and compare a chosen summary with the observed one.

5 Compare models

Evidence is the prior average of the likelihood. It is constant in the parameter for one model but matters across models. $B_{10} = p(D \mid H_1)/p(D \mid H_0)$ compares the two evidences; use proper priors and state their bounds. An excellent best fit may give weak evidence if only a tiny fraction of the prior fits well. Posterior odds are Bayes factor times prior odds. For two models with equal prior odds, $P(H_1 \mid D) = B_{10}/(1 + B_{10})$.

A noise-only p-value instead asks how often a test statistic at least as extreme would occur under H_0. In 5A the parameter posterior assumes a pulse, the Bayes factor compares models, and the noise-only test calibrates the search.